

Eg:-

```

int x=10;
{
    int y=20;
}

```

i) Stack-Based Allocation:-

- Memory is managed using a runtime stack.
- Each block/function creates an activation record.

ii) Activation Record with Links:-

- Each block contains:-
 - Local variables
 - Control link (to caller).
 - Access link (important).

Static (Access) link:-

- Used to access variables in outer (enclosing) blocks.
- Maintains lexical (static) scope.

Feature	Block-Structured Allocation	Simple Stack Allocation
• Scope	Nested scopes supported.	Only function-level.
• Access to outer variables	Yes (via access link).	No.
• Links used	Control + Access links.	Only control link.
• Flexibility	High.	Limited.
• Language support	C, Java, Pascal.	Basic languages.

8) A compiler detects errors in different phases. The three main types are:-